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Activity - 6

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(a) Rule: inflow = outflow at every node.

* Direct → Home = 1,913 - 1,146 - 402 = 365
* Push → Home = 1,037 - 365 - 438 = 234
* Push → Politics = 1,312 - 402 - 462 = 448

(b) (i) Conversion rate = 1346 / 4250 = 31.7%.

(ii) Largest flow = Exit site → Not subscribed, 1,043

(iii) Home → Read second article = 240 / 1037 = 23.1%.

(c) Node order: sort each column by size, largest at the top, to reduce link crossings.

Colour: use green for the path to subscribed and grey for the path to Not subscribed.

Labels: Show the count and % on each node, and label only flows of about 200 visits or more.

Highlight: Emphasise Exit → Not subscribed (the biggest loss) and open subscribe box X

(d) sports = 1,901 / 200 = 9.51cm

Home = 1,037 / 200 = 5.19cm

Politics = 1312 / 200 = 6.56cm

(e) R program

```
library(networkD3)

nodes <- data.frame(name=c("Direct visit", "Search engine"))

links <- data.frame(source=c(0,0,1,1),
                    target=c(3,4,3,4),
                    value=c(1146, 365, 234))

sankeyNetwork(Links = links, source = "source", Target="target",
              NodeID="name")
```

(f) Too many nodes or flows: Many thin, crossing links make it unreadable.

Comparing exact values or rates: link widths are hard to compare precisely. use a bar chart instead.